C30 Amino Acids, Proteins and DNA – Knowledge Organiser: Self-Test

AQA A-level Chemistry • Set 1: Amino Acids and Proteins **Paper 2**

#	Question	Attempt 1	Attempt 2	Attempt 3	Answer
1	Functional groups in amino acids				$-NH_2$ and –COOH
2	Zwitterion				Ion with both + and − charges
3	Amino acid in acidic solution				$-NH_3^+$ form
4	Amino acid in alkaline solution				$-COO^-$ form
5	Why amino acids have high melting points				Ionic attraction between zwitterions
6	Peptide link				–CONH–
7	Primary structure				Sequence of amino acids
8	Secondary structure				α-helix or β-pleated sheet (hydrogen bonds)
9	Tertiary structure				3D folding (H bonds, ionic bonds, disulfide bridges)
10	Conditions to hydrolyse a protein				6 mol dm^{-3} HCl, reflux

Fold the Answer column behind the page. Answer each question from memory, then unfold and check. Repeat on a later day in the next Attempt column.

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AQA A-level Chemistry • Set 2: Enzymes and Chromatography **Paper 2**

#	Question	Attempt 1	Attempt 2	Attempt 3	Answer
1	Amino acid that forms disulfide bridges				Cysteine
2	Disulfide bridge				S–S bond
3	Achiral amino acid				Glycine
4	Enzyme active site				Stereospecific
5	Enzyme inhibitor				Binds to the active site, blocking the substrate
6	Use of computer modelling in drug design				Designs molecules to fit an active site
7	Locating agent for amino acids				Ninhydrin
8	R_f equation				Distance moved by spot ÷ distance moved by solvent
9	How amino acids are identified by TLC				Compare R_f values
10	Bond broken in protein hydrolysis				Peptide (amide) bond

Fold the Answer column behind the page. Answer each question from memory, then unfold and check. Repeat on a later day in the next Attempt column.

C30 Amino Acids, Proteins and DNA – Knowledge Organiser: Self-Test

AQA A-level Chemistry • Set 3: DNA **Paper 2**

#	Question	Attempt 1	Attempt 2	Attempt 3	Answer
1	Monomer of DNA				Nucleotide
2	Three parts of a nucleotide				Phosphate, 2-deoxyribose, base
3	Four bases in DNA				Adenine, cytosine, guanine, thymine
4	Complementary base pairs				A–T and C–G
5	Hydrogen bonds between A and T				2
6	Hydrogen bonds between C and G				3
7	Shape of DNA				Double helix
8	DNA backbone				Sugar–phosphate
9	How cisplatin works				Binds to N on guanine; stops DNA replication
10	Side effect of cisplatin				Also damages healthy cells

Fold the Answer column behind the page. Answer each question from memory, then unfold and check. Repeat on a later day in the next Attempt column.